

Campus IQ — GAP · Readiness Check

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Target roles: Software Engineering Intern · Computer Engineering Intern · Embedded Systems Intern

Readiness Snapshot

You're a freshman, so the honest read is: long runway, early stage, and the gaps below are expected rather than alarming. The point of this check isn't to rank you against a senior — it's to show you what to build now so the internships are reachable when you apply.

Readiness level: Early — foundation in place, evidence not yet built.

Where You Stand Strong

- **A real (if early) team engineering experience.** Your work on the Texas A&M Engineering Project Team gives you something most freshmen don't have — documented collaboration on an engineering deliverable.
 - **The right starting foundation.** Python and engineering-computation basics from ENGR 102 are exactly where a future embedded/software intern should begin.
 - **Coherent direction.** Your interests, intended major, and target roles all point the same way. That focus is itself an asset.
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Must-Close Gaps

These are the things standing between you and a competitive internship application — in priority order.

1. **Demonstrated, documented projects.** This is the big one. You have coursework, but not yet a project you can *show* — with a README, tested code, and a written account of your process. For these roles, employers read your documentation as direct evidence you can do the work.
 2. **Depth beyond basics.** “Python basics” and “MATLAB basics” need to become demonstrated proficiency over the next two years — proven through projects, not just listed on courses.
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Nice-to-Have (Not Urgent as a Freshman)

- **Certifications.** You have none yet, and that’s fine at this stage. Don’t rush these — projects matter more right now.
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Your Runway

Graduation is May 2029. That’s a long, forgiving timeline — you can close every gap above *if* you start building documented project evidence now rather than waiting until junior year. The students who struggle aren’t the ones with gaps as freshmen; they’re the ones who wait until applications are due to start closing them.

Recommended Next Steps

1. **Turn one existing class project into a documented portfolio piece.** Take your ENGR 102 computation work, add a proper README, clean tested code, and a short write-up of your process. This single action closes your #1 gap.
 2. **Document as you go, everywhere.** In code, comment your logic and write testing notes. This builds the exact habit employers hire on — and it’s already what your professors are nudging you toward.
 3. **Target one technical internship or research assistantship for next summer.** Even an on-campus lab role adds the applied experience that ranks highest with employers.
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*Data routing note: Ethan's target roles (Software Engineering / Computer Engineering / Embedded Systems Intern) have no clean O*NET-SOC family match, so GAP resolves him to **web-search-only** per the resolution cascade — must-have vs. nice-to-have is derived from live DFW posting requirements, not O*NET importance scores. (O*NET-scored GAP applies to students whose target roles match one of the 10 mapped role families.)*